WORKSHEET

Chapter 8: Probability density functions

1 Which of the following are probability density functions?

a $f(x) = \frac{1}{2}(x^2 - 1)$ for $0 < x < 1$

b $f(x) = \frac{1}{2}$ for $2 < x < 4$

c $f(x) = \frac{x}{4}$ for $1 < x < 3$

d $f(x) = \frac{1}{12}(2x + 1)$ for $0 < x < 3$

e $f(x) = \frac{1}{2}(4x - 1)$ for $0 < x < 2$

2 Show that $f(x) = \frac{1}{(1 + x)^2}$ for $x \geq 0$ is a probability distribution function.

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3 Consider the function:

$$f(x) = \begin{cases} \frac{x^3}{5000}(10 - x) & \text{for } 0 \leq x \leq 10 \\ 0 & \text{for all other values of } x \end{cases}$$

a Show that $f(x)$ is a probability distribution function.

b Find $P(1 \leq X \leq 5)$.

c Find $P(X \geq 4)$.

4 For what constant, k, is $f(x) = ke^{-x}$ a probability density function on $[0, 1]$?

5 Consider the continuous random variable whose probability density function is $f(x) = \frac{x^3}{4}$ for $0 \leq x \leq k$.

Find the value of k.

6 Consider the continuous random variable, X, whose probability density function is $f(x) = 3x^2$ for $0 < x < 1$. What is the cumulative distribution function $F(x)$ for the variable?

7 Consider the continuous random variable, X, whose probability density function is $f(x) = \frac{x^3}{4}$ for $0 < x < 2$. What is the cumulative distribution function $F(x)$ for the variable?

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8 Consider the continuous random variable, X, whose probability density function is:

$$f(x) = \begin{cases} x + 1 & \text{for } -1 < x < 0 \\ 1 - x & \text{for } 0 \leq x < 1 \end{cases}$$

What is the cumulative distribution function $F(x)$ for the variable?

9 The cumulative probability density function for a continuous random variable, X, is:

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ 0.1x - 0.0025x^2 & \text{for } 0 \leq x \leq 20 \\ 1 & \text{for } x > 20 \end{cases}$$

Calculate the probability that $0 \leq x \leq 15$.

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Answers

1 **b**, **c** and **d** are probability density functions.

2 $\int_0^{\infty} \frac{1}{(1+x)^2} dx = 1$

3 **a** $\int_0^{10} \frac{x^3}{5000}(10-5)dx = 1$

b 0.1870

c 0.9130

4 $k \approx 1.582$

5 $k = 2$

6 $F(x) = \begin{cases} 0 & \text{for } x \leq 0 \\ x^3 & \text{for } 0 < x < 1 \\ 1 & \text{for } x \geq 1 \end{cases}$

7 $F(x) = \begin{cases} 0 & \text{for } x \leq 0 \\ \frac{x^4}{16} & \text{for } 0 < x < 2 \\ 1 & \text{for } x \geq 2 \end{cases}$

8 $F(x) = \begin{cases} 0 & \text{for } x \leq -1 \\ \frac{1}{2}(x+1)^2 & \text{for } -1 < x \leq 0 \\ 1 - \frac{1}{2}(1-x)^2 & \text{for } 0 \leq x < 1 \\ 1 & \text{for } x \geq 1 \end{cases}$

9 0.9375

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